

■ Day 2: Linux Commands

Complete Guide & Detailed Notes

■ Learning Objectives

By the end of this session, you should be able to:

- Understand the purpose and usage of essential Linux commands
- Navigate, create, manipulate, and delete files and directories
- Use **man** to explore new commands independently
- Build muscle memory through hands-on practice
- Understand file permissions, hidden files, and relative/absolute paths

■ Theory & Concepts

1. The Linux Filesystem Hierarchy

Linux organizes files in a single rooted tree starting at / (root). Your current location is called the **working directory**.

Directory	Purpose
/home/	User home directories
/etc/	System configuration files
/var/	Variable data (logs, caches)
/tmp/	Temporary files (cleared on reboot)
/bin/, /sbin/	Essential system binaries

2. Paths

- **Absolute path:** Starts from root (/), e.g. /home/user/embedded_projects
- **Relative path:** Starts from current directory, e.g. ./projects or ../scripts

Special directories:

- . — current directory
- .. — parent directory
- ~ — home directory of the current user

3. File Types in Linux

- Regular files (-)
- Directories (d)
- Symbolic links (l)
- Device files (b, c)
- Sockets and pipes

4. File Permissions (Brief Intro)

Each file has an **Owner**, **Group**, and **Others**, with permissions r (read), w (write), x (execute).

Example: -rw-r--r-- means the owner can read/write, while group and others can only read.

5. Hidden Files

Files starting with a dot (.bashrc, .gitconfig) are hidden. Use `ls -a` to see them.

6. Manual Pages (man)

- man is the built-in documentation system
- Sections: 1 (commands), 2 (system calls), 3 (library functions), etc.
- Navigation: q to quit, space to page down, / to search

■ Essential Commands – Deep Dive

Command	Syntax	Description	Examples
ls	ls [options] [path]	List directory contents	ls -la, ls -ltr
cd	cd [path]	Change directory	cd ~/proj, cd .., cd -
pwd	pwd	Print working directory	Shows current path
mkdir	mkdir [options] dir	Create directory	mkdir -p a/b/c
touch	touch file	Create file / update timestamp	touch main.c
rm	rm [options] file	Remove files/directories	rm -rf dir/
cp	cp [options] src dest	Copy files/directories	cp -r src/ dest/
mv	mv source dest	Move or rename	mv old.c new.c
cat	cat file	View file content	cat /etc/os-release
man	man command	Display manual page	man ls

■ Hands-on Practice – Step-by-Step

■ **Setup (from Day 1):** Make sure your Ubuntu VM or WSL2 is running and you are logged in.

Task 1: Practice Navigation

```
pwd # Check current location
cd / # Go to root
ls -la # List all files in root
cd ~ # Go home
cd embedded_projects/ # Go to your project folder
pwd # Confirm path
```

Task 2: Create and Organize Directories

```
cd ~/embedded_projects
mkdir -p c_programs linux_scripts projects
ls -l # Verify directories
```

Task 3: Create and Manipulate Files

```
cd c_programs
touch hello.c main.c utils.c
ls -la
cp hello.c hello_backup.c # Copy file
mv utils.c utilities.c # Rename file
rm main.c # Delete file
```

Task 4: Work with Hidden Files and Parent Directory

```
touch .secret_config # Create hidden file
ls -la # See hidden files
cd ../projects # Move to projects
touch file1.txt file2.txt
cd .. # Go back to embedded_projects
mv projects/file1.txt linux_scripts/
```

Task 5: Use cat and man

```
cat /etc/os-release # View OS info
man ls # Read manual
man cp # Learn more options
```

Task 6: Explore New Commands (at least 5)

Use man to learn these and practice:

- head — show first few lines of a file
- tail — show last few lines
- wc — word count
- find — search for files
- grep — search inside files

```
head -n 5 /etc/passwd
tail -n 10 /var/log/syslog
wc -l hello.c
find . -name "*.c"
grep "main" hello.c
```

■ Revision & Research

■ Research Question

"Why is Linux preferred for embedded systems?"

- **Open-source:** Free to modify and distribute
- **Modularity:** Can strip down to minimal size (e.g. using BusyBox)
- **Real-time support:** With PREEMPT_RT patches
- **Hardware support:** Wide range of architectures (ARM, x86, RISC-V)
- **Networking stack:** Robust and mature
- **Toolchain availability:** GCC, GDB, Make, etc.
- **Community support:** Massive ecosystem and documentation

■ Additional Research Topics for Day 2

- 1 What is the difference between `rm` and `rmdir`?
- 2 How do file permissions affect command behavior?
- 3 What are symbolic links and how do you create them (`ln -s`)?
- 4 Explore the `/proc` filesystem — why is it virtual?

■ Quick Reference Card

```
# Navigation
pwd
cd ~ / .. / -
ls -la / -ltr / -R

# Files & Directories
mkdir -p path
touch file
cp -r source dest
mv source dest
rm -rf dir

# View & Search
cat file
head / tail -n N
grep pattern file
find . -name "*.ext"

# Help
man command
command --help
```

■ Self-Assessment

Test yourself with these questions:

- 1 How do you go to your home directory from anywhere?
- 2 What command lists all files including hidden ones?
- 3 How do you create a directory and its parent directories in one command?
- 4 What is the difference between `cp file1 file2` and `mv file1 file2`?
- 5 How do you view the last 20 lines of a log file?
- 6 How do you search for all `.c` files inside `~/embedded_projects`?
- 7 What does `man -k network` do?